

UCS510: COMPUTER ARCHITECTURE AND ORGANIZATION

L	T	P	Cr
3	0	0	3.0

Course Objectives: Focus is on the architecture and organization of the basic computer modules viz. controls unit, central processing unit, input-output organization and memory unit.

Basics of Computer Architecture: Number System and code conversion , Logic gates, Flip flops, Registers, Multiplexer, De-multiplexer, Decoder, Encoder etc. IEEE 754 Floating point representation. 32bit/64bit

Register Transfer and Micro operations: Register transfer Language, Register transfer, Bus & memory transfer, Arithmetic micro operations, Logic micro operations, Shift micro operations, Design of ALU (**Decent work and economic growth**). Three state buffer, Binary Adder, Binary Incrementor.

Basic Computer Organization: Instruction codes, Computer instructions, Timing & control, Instruction Cycles, Memory, register, and input-output reference instructions, Interrupts, Complete computer description & design of basic computer. Direct and Indirect Address.

Central Processing Unit: General register organization, Stack organization, Instruction format, Addressing modes, Data transfer & manipulation, Program control, RISC, CISC. Register and memory stack, software and hardware interrupt, Green Computing (**Affordable and clean energy**).

Pipelining and Computer Arithmetic: Addition & Subtraction, Multiplication Algorithms, Division algorithms. Instruction Pipeline, Data Pipeline, Risk Pipeline (**Industry, innovation and infrastructure**). Dependencies in a pipeline processor, pipeline hazard.

Memory Unit: Memory hierarchy, Processor vs. memory speed, High-speed memories, Main Memory, Cache memory, Associative memory, Interleaving, Virtual memory, Memory management techniques. Direct Mapping, Set Associative Mapping.

Multiprocessors: Characteristics of multiprocessors, Interconnection structures (**Industry, innovation and infrastructure**), Inter-processor arbitration, Inter-processor communication & synchronization. Peripheral devices, I/O interface Data transfer schemes, Program control, Synchronous and asynchronous data transfer, Interrupt, DMA transfer, I/O processor.

Course Learning Outcomes (CLOs) / Course Objectives (COs):

On completion of this course, the students will be able to

1. Illustrate various elementary concepts of computer architecture including, syntax of register

transfer language, micro operations, instruction cycle, and control unit

2. Describe the design of basic computer with instruction formats & addressing modes
3. Explore various memory management techniques and algorithms for performing addition, subtraction and division etc
4. Interpret Concepts of pipelining, hazards, memory hierarchy, cache management and virtual memory.

Text Books:

1. Mano, Morris M., Computer System Architecture, Prentice Hall (1991) 3rd ed.
2. Hayes, J.P., Computer Architecture and Organization, McGraw Hill (1998) 3rd ed.
3. William Stallings, Computer Organization and Architecture, Pearson (2018), 11th ed.

Reference Books:

1. Hennessy, J.L., Patterson, D.A, and Goldberg, D., Computer Architecture A Quantitative Approach, Pearson Education Asia (2006) 4th ed.
2. Leigh, W.E. and Ali, D.L., System Architecture: software and hardware concepts, South Wester Publishing Co. (2000) 2nd ed.